



Spatial Truth: Location Service for Physical AI

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Connectcast Telco Announces "Spatial Truth": A Software-Defined Location Service to End Urban and Rural Navigation Blind Spots.

Amazon and Toyota achieve centimeter-accurate autonomy via software-defined network intelligence - zero hardware required.

Today, Connectcast announced **Spatial Truth**, a software-defined **Network-as-a-Sensor** location service that gives enterprise vehicle fleets **centimeter-level accuracy** with zero additional hardware. Spatial Truth is built for large-scale operators, such as Amazon and Toyota, that require reliable, lane-level accuracy across warehouses, ports, yards, logistics hubs, and public roadways.

The service eliminates location blind spots. In warehouses and ports, vehicles experience constant signal bounce-offs from steel structures, containers, cranes, and stacked inventory, creating large multipath errors that distort positioning. In dense urban environments, skyscrapers, glass façades, and metal infrastructure cause GPS reflections and interference that routinely generate 10-20-meter inaccuracies. And across rural open-road routes, vehicles and autonomous systems encounter complete signal dropouts, leaving navigation systems unable to determine even coarse location.

Spatial Truth resolves these blind spots by shifting accuracy from the vehicle to the network. Using only the standard radios such as 5G, Wi-Fi RTT, and terrestrial signals already built into today's connected vehicles. Spatial Truth fuses these inputs inside the 5G core to create a software-defined "source of truth" for location. The result is reliable, centimeter-level accuracy delivered at fleet scale, with zero hardware.

Today, approximately 72 million connected vehicles operate under conditions where GPS becomes unreliable or unusable. Warehouses, ports, and dense city centers produce multipath distortion; glass and steel surfaces generate false reflections; and rural areas experience full GPS signal loss. These errors cause lane-level navigation errors, missed ETAs, routing mistakes, slowed automation, and safety exposures, for fleets at scale. Of these **72 million** vehicles, we focus on a serviceable market of **45 million** already on 5G networks, where blind-spots drive **\$3.24B** annual impact.

Across enterprise fleets, these location errors collectively generate an estimated \$3.24B annual P&L drag. Historical solutions, such as adding LiDAR, RTK modules, or other high-cost sensors, cost **\$5,000+ per vehicle**, making autonomy expensive to deploy widely.

EXECUTIVE PERSPECTIVE "For years, connectivity and location were treated as silos. Spatial Truth proves the network *is* the sensor," said the VP of IoT Engineering at Cisco. "By enabling this through the Cisco IoT Control Center, the #1 platform for ~300 million connected devices, *we stop selling megabits and start selling guaranteed centimeter-level location.*"

90-DAY ACTION PLAN:

Proposed plan for the enterprise to validate the proof of concept:

1. Days 1-30: Audit current LMF/Core architecture and align with Toyota and Amazon on dead-zone pain points.
2. Days 31-60: Execute a PoC using existing gNB hardware to demonstrate centimeter accuracy in a simulated signal bounce-off environments.
3. Days 61-90: Launch Location Service beta tier, converting connectivity into high-margin SaaS location service.

Sources

Statista – [Connected Cars Worldwide : 400M+ vehicles projected by 2025](#)

5G Automotive Association (5GAA) – [Over 300M vehicles with advanced connectivity today](#)

Juniper Research/ Business Standard – [Commercial vehicle ~20% of connected vehicle base by 2027.](#)

Internal FAQs

Q 1) What is the customer demand and market opportunity for Location Service?

(Appendix A: Market Projections and SAM)

A: In consumer space the success of Life360 and its 80 million subscribers proves: people pay for location accuracy.

Consumers already expect continuous location accuracy, even when the network fails. That's why Apple built Emergency SOS, and why Skylo provides NTN continuity in rural dead zones. These technologies didn't *optimize* uncertainty—they removed it.

Spatial Truth brings that same leap to the enterprise: It eliminates the root problem, not the symptoms. Without it, commercial vehicle fleets keep paying the "location uncertainty tax"—every hour, every route, every day.

We define the market not by data usage, but by the active connected vehicle base. Our Serviceable Addressable Market (SAM) consists of **45M** commercial vehicles globally that currently pay a "location uncertainty tax."

- The opportunity at a **\$6/month blended ARPU**, is **\$3.24B**
- The annual projection: With a conservative 5% penetration of this connected vehicle base (2.25M vehicles X \$6/month X 12 months), Spatial Truth projects \$162M in Annual Recurring Revenue (ARR).
- The Demand: Much like Life360 proved consumers will pay for location certainty, enterprises like Amazon and Toyota require a centimeter-accuracy SLA to scale operation efficiency. Spatial Truth provides this through three subscription tiers:
 - \$3/month for <1m (Bronze)
 - \$7/month for <30cm (Silver)
 - \$15/month for <10cm (Gold)
- Exclusions for Conservatism: The 45M figure excludes consumer passenger vehicles (despite Life360's 80M+ subscriber proof-of-concept), emerging 5G markets, and non-automotive sectors like maritime or agriculture.

Q 2) Why is the monthly subscription of, "\$3 / \$7 / \$15," per month the right price point?

(Appendix B: Pricing and Financial Projection Deep Dive)

A: The \$6.00 blended monthly ARPU represents a shift from selling low-margin "megabits" to high-margin Location Service. For a truck generating \$200k/year in revenue, a \$15/month (\$180/year) "Gold" tier represents just 0.09% of operating costs. Payback Calculation: If our location accuracy prevents just one 25-minute warehouse delay per month, the subscription pays for itself (\$204 in annual recovered time vs. \$180 annual cost if priced at \$15/month):

- 25 min delay = ~0.42 hours.

- \$40/hour burdened driver cost = 0.42 hours X \$40/hour = \$17 per incident/month.
- 12 months X \$17 incidents/month = \$204 per vehicle in annual waste.

Conclusion: A single avoided urban city missed turn or warehouse dwell incident per month pays the entire annual investment.

Q3) How does this impact the P&L of a 1,000-vehicle enterprise fleet? (*Appendix C: Financial Projection for 1000-fleet vehicle*)

A: Location error is a direct P&L drag. For a 1,000-vehicle fleet generating \$180M–\$200M in revenue, Spatial Truth recovers \$11M–\$29M annually in location uncertainty tax.

- \$1.2M–\$3.6M/year in routing efficiency by eliminating 1-3% GPS navigation loss.
- \$800K–\$1.5M/year in yard efficiency by reducing warehouse/yard inefficiency by 8-15 min./stop).
- 10-20% Reduction in insurance premiums by providing a certified “source of truth” for liability.

Table 1.0: Location uncertainty tax recovers \$11M - \$29M annually for 1,000-Vehicle Fleet

Use Case	Stakeholder	The Problem	The Impact	The Recovery
#1 Routing	Fleet Manager	10–20m drift causes missed turns and circling.	1-3% inefficiency; \$1.8M–\$5.4M gross loss.	\$1.2M–\$3.6M in direct waste recovered.
#2 Yard Dwell	Logistics Lead	Poor fidelity adds 8-15 mins of manual dwell per stop.	500 fleet-hours lost daily; \$6M+ annual waste.	\$800K–\$1.5M saved via automated check-ins.
#3 Utilization	COO	Unreliable telemetry prevents confident job reassignment.	5-12% utilization drag; ~\$9,000 lost per truck.	\$9M–\$21.6M utilization loss.
#4 Insurance	Safety Officer	GPS "jitter" creates ambiguous liability evidence.	10-20% premium hikes on \$6.5K–\$10K base costs.	\$650K–\$2M reduction in annual premiums.
#5 Autonomy	CTO	GPS drift forces disengagements and \$5K+ hardware fixes.	Frozen CAPEX; autonomy programs stuck in test loops.	Accelerated ROI; scales L2.5/L3 to urban routes.

Q 4) What real customer problem does Spatial Truth solves?

A: Enterprises currently pay a "location uncertainty tax." Location errors of 10-20 meters in urban cities, lane-level inaccuracies in warehouses, and rural signal dropouts lead to missed ETAs, routing errors, and safety exposures.

Spatial Truth solves this by addressing three critical friction points:

- Eliminates the "Integrator's Burden": Traditional accuracy requires thousands in upfront CAPEX per vehicle for hardware (LiDAR/RTK), followed by months of installation. Spatial Truth is a Zero-CAPEX software update that scales across global fleets instantly via the Cisco IoT Control Center.
- Predictable Automation: Spatial Truth moves location from being a "best-effort" utility to a high-accuracy sensor, enabling L2.5/L3 autonomy to function reliably in dense urban or indoor environments where GPS currently fails.
- The "Google Maps" Moment for Enterprise: Spatial Truth provides the same leap in certainty that consumers felt when trip-planning became effortless. The same "peace of mind" that Life360 provides families.

Q7) Who should build this?

A: The solution requires orchestration across four key layers:

- Infrastructure Providers (The Backbone): Companies like Cisco, Ericsson, and Huawei. They provide the 5G Standalone (SA) cores and Location Management Functions (LMF), where the AI layer is hosted. *The Advantage*: These players own the raw signal data, allowing them to use reflection signals to sharpen accuracy—a capability standalone GPS chips lack.
- Platform Orchestrators (The Scale): Platforms like the Cisco IoT Control Center. They provide access to the ~300 million connected devices.
- Global Constellations (The Continuity): Partners like Skylo provide the NTN (Non-Terrestrial Network) bridge for rural continuity.
- The Enterprise Fleets (The Demand): Large-scale operators like Amazon and Toyota that provide the high-volume, recurring revenue data.

Q 6) Why should the build? What leverage do they have?

A: Ownership of the raw network signal. Unlike standalone GPS, Spatial Truth uses reflections as "signal fingerprints" to sharpen accuracy where others see noise. Spatial Truth doesn't optimize uncertainty; it removes it.

Q 7) How does it work? (*Appendix D: Technical Overview*)

A: Spatial Truth delivers centimeter-level accuracy by combining AI-driven signal processing with seamless terrestrial satellite continuity inside the 5G network core.

- Super-Resolution AI (Physics to ROI): Spatial Truth hosts AI-powered Zainar as a containerized software module inside the 5G LMF (Location Management Function). Instead of treating reflections as noise, the AI interprets them as “fingerprints,” producing centimeter-level location accuracy.
- The Skylo Bridge: For rural continuity, Spatial Truth utilizes NB-NTN standards to trigger a seamless handover from 5G to GEO satellites via the S8 interface, ensuring location visibility in rural areas.
- Deep Integrated Positioning: Spatial Truth network blends 5G and Wi-Fi RTT (802.11mc passive ranging) at the control plane, ensuring a drone navigates safely indoors, around building, or on balconies, without requiring private Wi-Fi access.